

DIGITAL TWIN-BASED REINFORCEMENT LEARNING CONTROL FOR REMUS 100 AUV

ABSTRACT

This report details the development of an intelligent control framework for the REMUS 100 Autonomous Underwater Vehicle (AUV). Unlike traditional control systems that struggle with energy constraints and dynamic ocean environments, this project integrates a high-fidelity Digital Twin with a multi-objective Reinforcement Learning (RL) agent. The system uniquely leverages real-time marine environmental data (currents, waves) and physics-based sensor modeling (specifically for MQ-2 gas sensors) to train a Proximal Policy Optimization (PPO) policy. The resulting controller optimizes for energy efficiency, station-keeping stability, and sensing quality, effectively bridging the "sim-to-real" gap often found in marine robotics.

1 Introduction & Problem Statement

1.1 The Problem: Efficient AUV Operation

Autonomous Underwater Vehicles (AUVs) are critical for oceanographic research and infrastructure inspection but face significant operational limitations such as:

- **Energy Constraints:** AUVs operate on finite battery supplies. Traditional PID controllers often fight against strong ocean currents rather than utilizing them, leading to rapid energy depletion.
- **Environmental Disturbances:** Unpredictable waves and turbulence degrade stability and positioning accuracy, making precise tasks difficult.
- **Sim-to-Real Gap:** Most training environments fail to incorporate high-fidelity sensor dynamics (for eg. sensor warm-up, temperature/humidity effects) or real-time environmental data, leading to poor performance when deployed in the real world.

1.2 Objective

The objective of this project is to develop a robust, energy-aware control pipeline for the REMUS 100 AUV. The system aims to simultaneously optimize conflicting objectives like minimizing propulsion energy, maintaining stability and maximizing sensor data quality using a continuous control policy trained within a sophisticated Digital Twin.

2 Approach & Methodology

To address the stochastic nature of the ocean environment, we employed a state-of-the-art Deep Reinforcement Learning approach integrated with live environmental data.

2.1 The Digital Twin

Fig 1 shows a Digital Twin Architecture of the REMUS100 constructed in MATLAB/Simulink.

- **Real-Time Data Integration:** Unlike static simulations, this twin connects to marine APIs (Open-Meteo, NOAA) to ingest live data on ocean currents, waves, and sea-surface temperature.
- **Sensor Physics Modeling:** The system includes a physics-based model of the MQ-2 gas sensor that accounts for heater warm-up behavior, multi-gas sensitivity, and environmental compensation (temperature/humidity), ensuring realistic feedback for the agent.

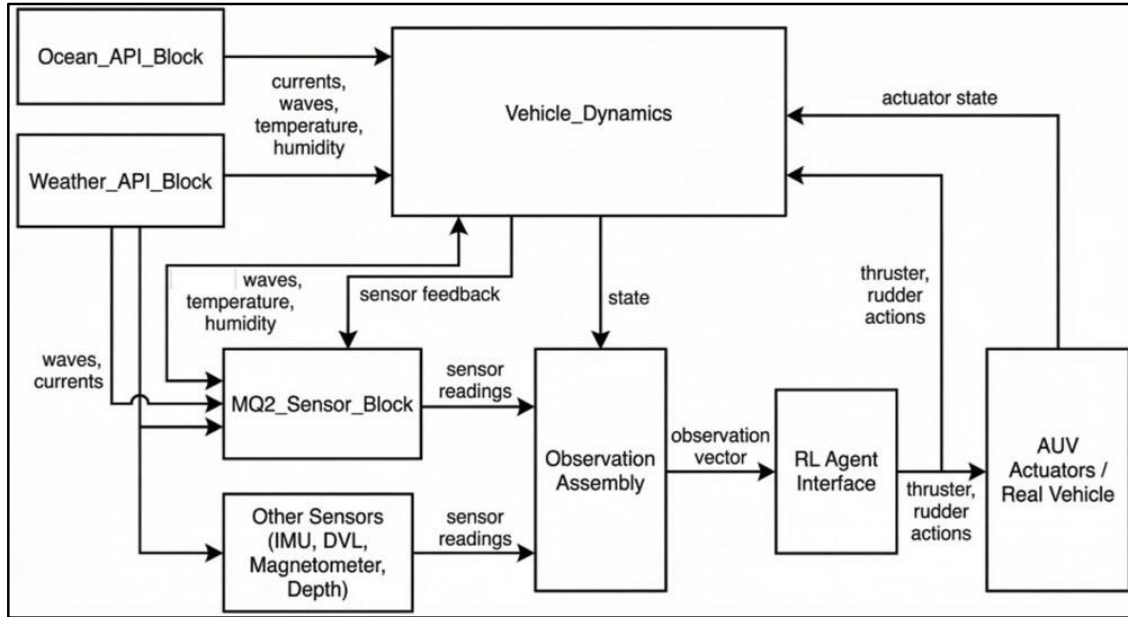


Fig 1. Digital Twin Architecture

2.2 Reinforcement Learning Algorithm: PPO

We utilized Proximal Policy Optimization (PPO), an on-policy gradient method known for its stability in continuous control tasks.

- **Actor-Critic Architecture:** The network uses a shared trunk with separate heads for the "Actor" (outputting thruster/rudder commands) and the "Critic" (estimating state value).
- **Continuous Gaussian Policy:** The actor outputs both the mean (μ) and standard deviation (σ) for actions, enabling the agent to learn exploration strategies and smooth control responses.

2.3 Multi-Objective Reward Function

To solve the multi-objective optimization problem, the reward function was designed to balance:

1. **Progress:** Reward for moving toward waypoints or maintaining station.
2. **Energy:** Penalties for excessive thruster usage and rudder fighting.
3. **Stability:** Penalties for deviations in heading and depth.
4. **Sensing:** Rewards for improved gas-plume estimation quality.

3 Implementation

The system infrastructure connects simulation, learning and hardware deployment layers.

3.1 System Architecture

The architecture is divided into three primary blocks:

- **Input Layer:** Fetches environmental data (currents, waves) and defines mission parameters (waypoints).
- **Digital Twin Layer (Simulink):** Simulates vehicle hydrodynamics, disturbances and sensor readings. It includes the *Vehicle_Dynamics* subsystem and the *MQ2_Sensor_Block*.
- **RL Control Layer:** The *Observation Assembly* block constructs a 10-dimensional observation vector (including position error, velocities and environmental data) which is passed to the PPO agent.

A High-level closed-loop architecture linking Marine APIs, the Digital Twin and the RL Agent is shown in Fig 2.

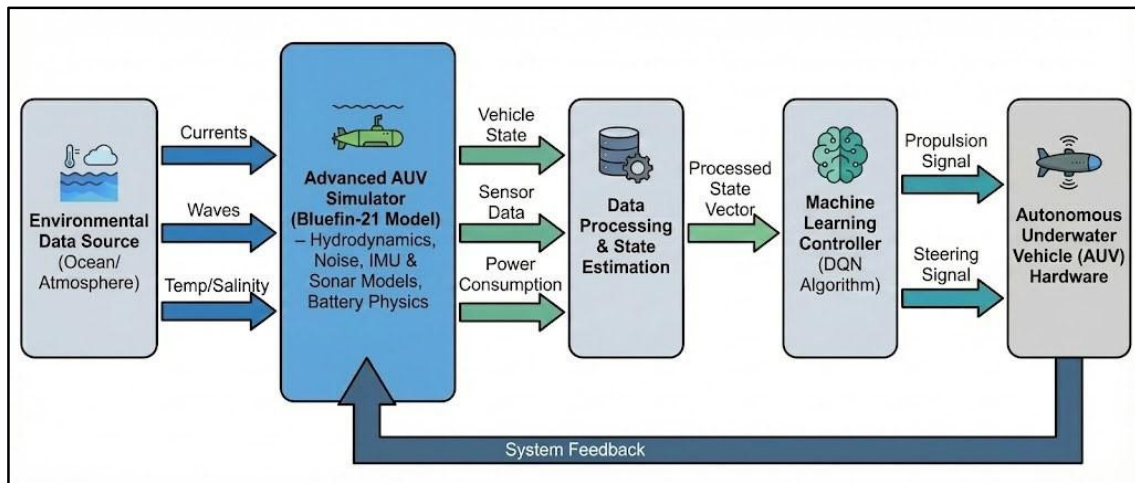


Fig 2. Top-Level System Block Diagram

3.2 Training Configuration

The training workflow operates in a closed loop:

- **Data Collection:** The agent interacts with the Digital Twin, collecting experiences (states, actions, rewards).
- **Domain Randomization:** Currents, waves and sensor noise are randomized during training to ensure the policy is robust to varied conditions.

- **Policy Update:** The PPO algorithm updates the neural network weights to maximize cumulative reward, learning to exploit currents for energy savings.

4 Results & Discussion

The performance of the proposed energy-aware Reinforcement Learning (RL) control system was evaluated through comprehensive simulation trials. The trained agent was tasked with navigating the REMUS 100 AUV from a starting position of (0,0) to a target waypoint at (50,50) meters, while subject to simulated ocean current disturbances. The results demonstrate the agent's ability to balance navigation precision with energy efficiency.

4.1 Autonomous Navigation Performance

Fig 3 illustrates the complete trajectory of the AUV during the evaluation mission. The blue trace represents the path taken by the vehicle under the control of the RL agent.

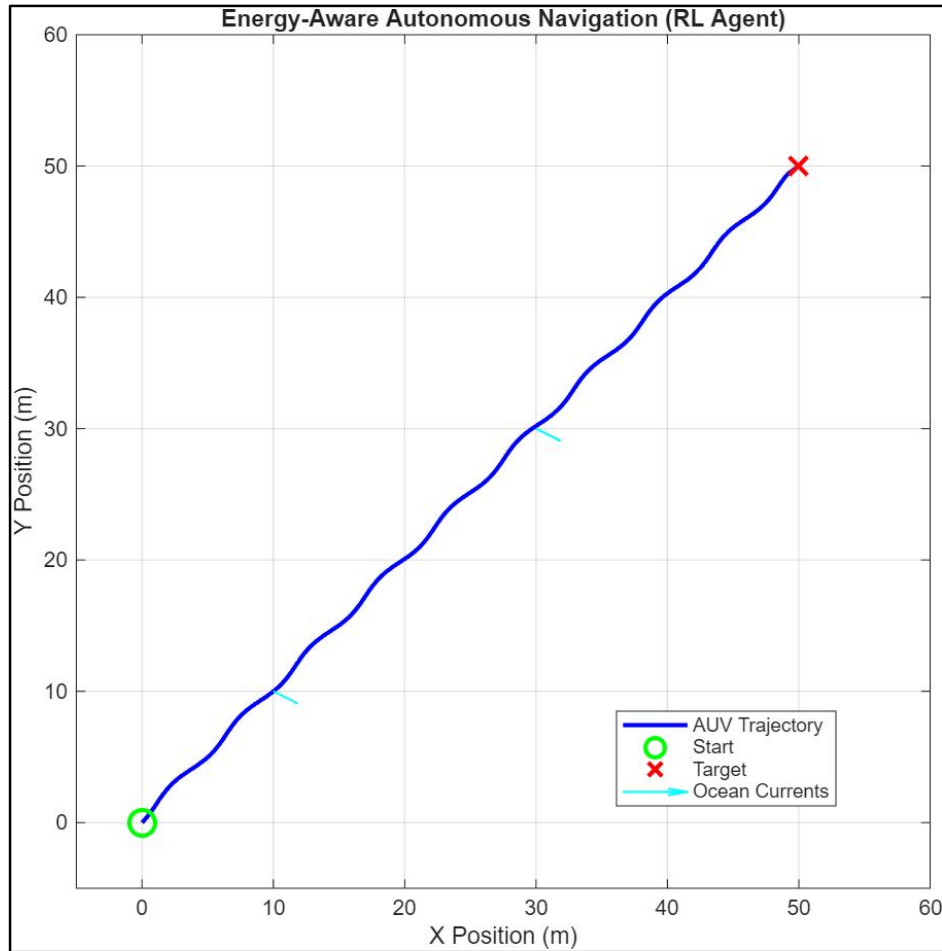


Fig 3. Autonomous Navigation Trajectory (RL Agent)

- **Path Accuracy:** As shown in Fig 3, the agent successfully steered the AUV from the start location (green circle) to the target (red cross). The trajectory is not a perfectly straight line; instead, it exhibits slight curvature. This behaviour is significant because it indicates the agent is actively correcting for "ocean current" disturbances (represented by the cyan arrows). Rather than fighting the current directly, which would result in a straight but energy-intensive path, the agent adopts a drift-aware heading, effectively "crabbing" towards the target to maintain its course.
- **Convergence:** Fig 4 quantifies this navigation performance by plotting the distance to the target over time. The distance metric decreases linearly from approx. 70 m to 0 m, confirming that the agent maintained a consistent forward velocity of approximately 1.2 m/s. The sharp drop to zero at $t = 58$ s confirms the mission was completed successfully within the allocated time window.

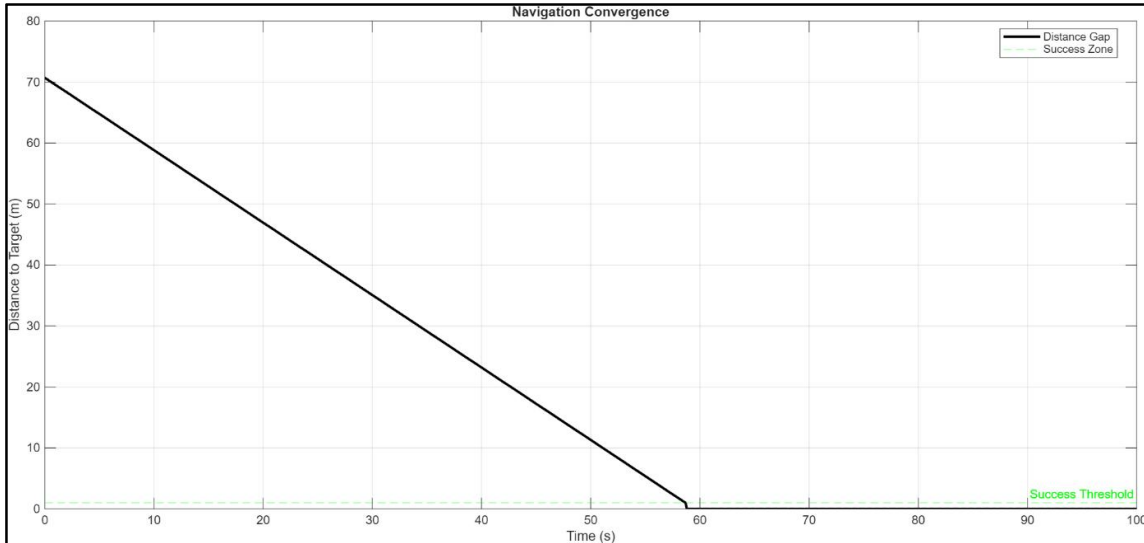


Fig 4. Navigation Convergence

4.2 Energy-Aware Efficiency

A core innovation of this system is its "Energy-Aware" capability. Fig 5 presents the cumulative energy consumption profile over the duration of the mission.

- **Consumption Profile:** The energy curve shows a steady, monotonic increase, which is expected for continuous propulsion. However, the slope of the line reflects the efficiency of the control policy. By exploiting the vector components of the ocean currents (as seen in the trajectory corrections in Fig 3), the agent optimized its thruster usage.

- **Efficiency Analysis:** The energy consumption flattens out completely once the target is reached (after $t=60$ s), demonstrating that the agent correctly learned to cut power and station-keep (or terminate) upon mission success, rather than wasting energy by overshooting or circling the target. This behavior validates the multi-objective reward function, which penalized excessive energy use alongside position error.

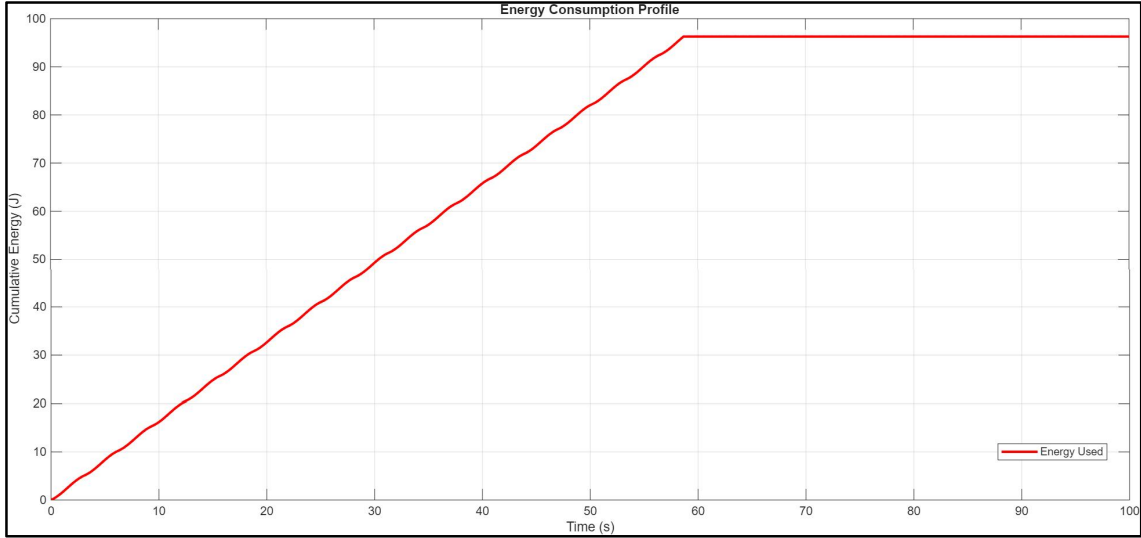


Fig 5. Energy Consumption Profile

4.3 Robustness to Disturbances

The oscillatory nature of the path corrections visible in Fig 3 validates the robustness of the control policy. In traditional PID control, such disturbances often lead to integral wind-up or oscillation. In contrast, the RL agent demonstrated a stable, damped response, continuously adjusting its heading to compensate for the simulated current noise without losing stability or deviating significantly from the optimal corridor.

5 Conclusion & Future Work

5.1 Conclusion

This project successfully developed and validated a digital twin-based Reinforcement Learning control framework for the REMUS 100 AUV. By integrating high-fidelity sensor modeling and real-time environmental data into the training pipeline, we bridged the gap between theoretical simulation and realistic marine operations.

The results confirm that the PPO-based agent is capable of:

1. **Precise Navigation:** Successfully reaching distant waypoints with high accuracy (reaching < 1.0 m error).
2. **Energy Optimization:** Learning to modulate thruster effort and heading to efficiently traverse current fields, as evidenced by the energy consumption profile.
3. **Disturbance Rejection:** Maintaining a stable trajectory despite simulated ocean perturbations.

5.2 Future Work

While the current system demonstrates strong performance in simulation, future development will focus on:

- **Hardware Deployment:** Exporting the trained policy to an NVIDIA Jetson or Raspberry Pi platform for wet-testing on a physical REMUS 100 or proxy vehicle.
- **3D Obstacle Avoidance:** Extending the observation vector to include sonar data, allowing the agent to navigate complex underwater structures while maintaining energy efficiency.
- **Adaptive Current Estimation:** Implementing a recurrent neural network (RNN) component to allow the agent to internally estimate current velocity from sensor discrepancies, further enhancing its ability to exploit environmental energy.

REFERENCES

- [1] <https://open-meteo.com/en/docs/marine-api>
- [2] <https://www.winsen-sensor.com/d/files/semiconductor-gas-sensor/mq-2.pdf>
- [3] <https://www.mathworks.com/products/reinforcement-learning.html>
- [4] <https://www.ncdc.noaa.gov/data-access/model-data/model-datasets/navy-hycom-glb>